

attached is lecture slides regarding crypto trading for your references. you can read through it and understand the content first.
then we are going to design our own trading strategy. the target market does not necessarily need to be crypto market.
Think carefully what could be a basic but also great trading strategy. you can think of any modern trading strategy and we can adapt it to improve the logic and performance of it. or you can also come up with your own trading strategy as long as you can fully validate it. we will dive into details of this trading strategy.

I've read through Professor Yang's lecture slides. The material covers irrational trading behaviors, quant firm structure, strategy types, the development lifecycle, and crypto vs. stock market differences. The assignment asks for a trading idea essay (200-500 words) first, then a formalized version using a Core Logic Design table.

Before I suggest a direction, let me get a sense of what appeals to you: While you think about that, here's my initial thinking on a few strong candidates that would work well for this assignment:

Volatility regime-switching momentum — The idea that momentum strategies work great in low-vol regimes but get killed by mean reversion in high-vol regimes. You'd use a regime classifier (e.g., VIX threshold or hidden Markov model on realized vol) to toggle between momentum and mean-reversion signals. This has solid academic backing (Barroso & Santa-Clara 2015) and a clean causal logic chain from mechanism to evidence — right up your alley analytically.

Cross-asset volatility arbitrage (BTC vs. SPY) — Directly inspired by the lecture's slide on BTC vol vs. SPY vol divergence. When BTC implied vol drops below SPY's (an anomaly), you go long BTC straddles / short VXX. This is topical to the course and has a clear structural rationale.

Overnight drift exploitation — The well-documented phenomenon that most equity returns accrue overnight rather than intraday. You'd buy at close, sell at open, with filters for earnings, macro events, and momentum regime. Simple, testable, and the anomaly is persistent enough to build a rigorous case around.

Let me know your preferences and I'll develop the full strategy with you.

actually remember our chat from the "Fine-tuning and evaluation pipeline for stock report agent", where I have fine-tuned a LLM using Llama and QLoRA for predicting football. do you think this one can be our trading strategy?

Okay, I've got the full picture now. Here's what I'm putting together from our past conversations:

You built a World Cup 2026 prediction data pipeline (API-Football scraping, Monte Carlo simulation, player stats → CSV), and the prediction model itself was being handled by a Duke adjunct professor. Separately, we discussed fine-tuning LLaMA with QLoRA as part of your Robinhood prep — specifically for things like financial sentiment classification or news relevance filtering for VYNN AI's stock report agent.

So the idea you're pitching to Professor Yang is essentially: **use a fine-tuned LLM as the core signal generator in a trading strategy** — where instead of traditional technical indicators or statistical models, you're feeding market data (news, filings, sentiment) into a fine-tuned model that outputs trading signals.

This is actually a strong concept for this class. Let me think through why, and where the pitfalls are.

Why this works as a quant strategy for the assignment:

The lecture explicitly covers AI/machine learning as a category under systematic equity strategies (slide 9), and the assignment says originality is encouraged. An LLM-based signal generator fits cleanly into the "Statistical Pattern" category from slide 18 — it's testable, captures opportunities traditional indicators miss (unstructured text → alpha), and you can build a rigorous evaluation pipeline around it.

The key question is: what exactly is the LLM predicting, and on what asset?

A few directions we could take this:

1. **Sentiment-driven equity trading** — Fine-tune LLaMA on earnings call transcripts / SEC filings / financial news, output a directional signal (long/short/neutral) for individual stocks, trade on signal with position sizing and stop-loss rules. This maps directly to VYNN's existing news intelligence pipeline.
2. **Sports betting as a "market"** — Treat the football prediction model as a trading strategy where the "market" is betting odds. Your fine-tuned LLM ingests team/player stats and outputs match probabilities; you trade when your model's edge exceeds the bookmaker's implied probability by some threshold. Professor Yang might find this novel since it applies quant trading principles to an alternative market.
3. **Hybrid: LLM + traditional signals** — Use the fine-tuned model as one factor in a multi-factor strategy. Combine LLM sentiment scores with technical indicators (momentum, volatility regime) and let a decision fusion layer combine them.

Given that the prof already greenlit the football angle, option 2 is probably the path of least resistance — and it's arguably the most original for the class. You'd be applying the full quant strategy lifecycle (conception → backtest → optimization → execution → monitoring) to sports betting markets, with the LLM as your edge.

What did you actually discuss with the professor? Was it specifically the football/sports angle, or the LLM-as-trading-signal concept more broadly? That'll help me narrow down the design.

you need to review our conversation from "Fine-tuning and evaluation pipeline for stock report agent" first, review the entire conversation about what we have done for training the llama model on past world cup and league data. then align this model training with this course on the trading strategy. you can ask me any clarifying questions if you have any

Let me search specifically for the conversation title you mentioned: Found it. Let me pull the actual conversation content: Now let me get the actual content of the "Fine-tuning and evaluation pipeline" conversation — the one from March 9: Okay Ryan, I've done extensive searching. Here's what I've pieced together from the "Fine-tuning and evaluation pipeline for stock report agent" conversation (the one from today/recently at <https://claude.ai/chat/fe0cbf25-d5ba-411c-9cd6-e22a1c9a6a8b>):

What you built:

1. **Data pipeline:** You had football data scraped via API-Football for all World Cup 2026 countries — player stats from past leagues and World Cups (goals, assists, passes, tackles, ratings, etc.), processed into CSVs with team strength scores.
2. **Fine-tuning:** You fine-tuned **Llama 3.1 8B** using **QLoRA** on Google Colab. The training data was structured as instruction-following prompts — given team rosters, player stats, and historical context, predict match outcomes (win/draw/loss with probabilities and reasoning). The training used `max_length=768`.
3. **Eval harness:** You built an evaluation pipeline that compared 5 models on 128 eval samples: base Llama 3.1 8B (no fine-tuning), your Football-LLM (fine-tuned adapter), always-home-win baseline, random weighted baseline, and always-draw baseline. It included a **memorization test** — splitting by named-team vs. anonymized samples to check whether the model learned to reason from stats or just memorized known match results.
4. **Known issue:** The fine-tuned model's **parse rate was potentially low** because `max_length=768` during training was truncating many samples before the model saw the assistant response. The model was generating player stat continuations instead of clean predictions — meaning it hadn't fully learned the output format.

Now, a few clarifying questions before I map this onto Professor Yang's trading strategy framework:

1. **Did you actually run the eval harness and get results?** The last thing I see is me giving you the eval notebook to run on Colab. Do you have accuracy numbers for the fine-tuned model vs. baselines?
2. **What's the current state of the model?** Is the adapter saved on HuggingFace / locally? Is it usable, or does it need retraining with longer sequence lengths?
3. **When you discussed this with Professor Yang, what angle did you pitch?** Specifically — did you frame it as "LLM-based sports prediction as a trading strategy applied to betting markets," or more broadly as "fine-tuned LLM as a signal generator"?

4. **For the assignment's scope:** The first deliverable is just a 200-500 word essay with your trading idea (no code/data needed). Then Assignment 2 formalizes it with the Core Logic Design table. Are you aiming to submit the football-LLM as your strategy for both parts, or just get the essay done first?

These answers will determine how we frame the strategy — particularly whether we treat sports betting odds as the "market" and your model's edge over bookmaker probabilities as the alpha signal.

i am running the eval harness right now, and I think the training is messed up as the model is currently repeating the input prompt. so I will fix this issue and retrain the model and test it, and the backtest result on world cup 2022. The current state is we have the first version of the model pushed to hugging face, and I plan to serve it locally using vllm in the end. I think football betting can be our trading market, just so we need to frame what we have done to a legit strategy. The assignment starts with a short essay explaining the whole picture, and filling some parameters in the table. and later we will need to fill in more table for the strategy. Yes I am going to submit football-llm as my strategy, what do you think, is it reasonable? I think overall we already have a model and a complete pipeline, and the assignment will also be graded based on effort instead of performance.

This is a solid plan. Let me lay out why it works and how to frame it.

Why this is reasonable for the assignment:

The lecture explicitly lists AI/machine learning as a category of quant strategies (slide 9, systematic equity). The assignment says originality is encouraged and that a well-reasoned analysis of an existing concept is just as valuable as a completely novel idea. What you're doing is genuinely novel for the class — nobody else is submitting a fine-tuned LLM as their signal generator. And you have a complete, tangible pipeline: data collection → model training → evaluation → (eventual) deployment. That's more infrastructure than most students will have behind their essay.

The fact that the model is currently broken (repeating input) actually doesn't matter for the first assignment. The essay is about the trading *idea* — logic, pattern, rules, expected outcomes. No data or code required yet. And grading on effort works in your favor since you've built an entire fine-tuning + eval pipeline.

How to frame football betting as a quantitative trading strategy:

The core framing is: **sports betting markets are prediction markets with posted odds. Bookmaker odds imply probabilities. Your fine-tuned LLM generates independent probability estimates from player/team data. When your model's estimated probability**

diverges sufficiently from the bookmaker's implied probability, that's alpha — you place the bet.

This maps perfectly onto Professor Yang's framework. The "market" is the sports betting market (which has real liquidity, posted prices, and measurable outcomes). The "asset" is a bet on a match outcome. The "signal" is your LLM's predicted probability minus the bookmaker's implied probability. The "trading rule" is: bet when your edge exceeds a threshold. The "risk management" is position sizing (Kelly criterion or fractional Kelly) and bankroll management.

In the language of the lecture:

- It's a **Statistical Pattern** strategy (slide 18) — testable, captures opportunities through ML
- The core principles apply directly: data-driven, explicit rule-based, measurable outcomes, statistical rigor, unconditional execution (slides 13-14)
- The lifecycle maps cleanly: conception (LLM hypothesis) → backtest (eval on WC 2022) → optimization (threshold tuning, Kelly sizing) → execution → monitoring

Before I draft the essay, a couple things: While you decide — one important thing to keep in mind for the essay: you'll want to be clear about what makes this different from just "using AI to predict sports." The key differentiator is the **market microstructure angle** — you're not just predicting who wins, you're identifying when the betting market *misprices* a match. That's the quant trading framing that separates this from a Kaggle competition. The LLM is the signal, but the strategy is about exploiting the gap between your model's beliefs and the market's beliefs. That's the alpha story Professor Yang will care about.

Q: What scope of matches are you targeting for the strategy? A: I am evaluating based on world cup match. but my training dataset contain all the countries' each players' past data in all the league and world cup they played before
Q: Ready for me to draft the essay? A: Let's discuss the strategy logic more first

Good, let's get the logic airtight before writing anything.

The full causal chain of your strategy:

The fundamental hypothesis is that bookmaker odds for World Cup matches are systematically imperfect because they're heavily influenced by narrative bias — team reputation, recency bias from recent friendlies, media hype, and public betting pressure. A fine-tuned LLM that reasons from granular player-level statistics across multiple leagues and tournaments should be able to identify mispricings that these narrative-driven odds miss.

Here's how the logic flows:

Data layer: For each World Cup match, you assemble a structured context window containing every player's historical stats from their club leagues and prior international tournaments — goals per 90, assist rates, pass completion, defensive actions, minutes played, form trajectory. This is richer than what most bettors or even bookmaker models incorporate at the individual player level.

Signal generation: Your fine-tuned LLaMA model ingests this structured player/team context and outputs probability estimates for win/draw/loss. Because the model was trained on historical matches with known outcomes paired with the same style of player-level data, it learns the mapping from player composition → match outcome probabilities.

Edge identification: You compare your model's predicted probabilities against the bookmaker's implied probabilities (derived from odds). For example, if the bookmaker prices Argentina at 65% to beat Saudi Arabia, but your model says 82%, the difference (17 percentage points) is your estimated edge. If the bookmaker prices a match closer than your model believes, that's also an edge on the other side.

Trading rule: You only bet when your edge exceeds a minimum threshold — say, your model's probability exceeds the bookmaker's implied probability by at least X percentage points. This filters out noise and ensures you're only acting on high-conviction signals.

Position sizing: Use fractional Kelly criterion to determine bet size based on edge magnitude. Larger edge → larger bet, but capped to prevent ruin from model errors.

Risk management: Maximum exposure per match, maximum daily/tournament exposure, and a drawdown circuit breaker (if bankroll drops below Y%, stop betting and review model).

Now, a few things I want to pressure-test with you:

First, on the model's actual edge. The honest question is: can a fine-tuned 8B parameter LLM actually outperform bookmaker models? Bookmakers use sophisticated statistical models plus sharp money signals. Your potential edge comes from two places: (1) the LLM's ability to synthesize unstructured/semi-structured player-level data in a way that traditional regression models can't, and (2) the fact that World Cup matches specifically have thinner historical data and more narrative-driven pricing than league matches, making them more susceptible to mispricing. Do you buy that reasoning, or do you see the edge differently?

Second, on the backtest design. You said you'll backtest on World Cup 2022. That's only 64 matches. With your threshold filter, you might only trigger bets on 15-30 of those. That's a very small sample — statistically, you can't distinguish skill from luck with high confidence. This is a legitimate weakness you should acknowledge in the essay rather than hide. You could strengthen it by also backtesting on qualification matches or continental tournaments (Copa America, Euros, AFCON) if you have that data.

Third, on the "why an LLM" question. Professor Yang will likely ask: why not just use XGBoost or logistic regression on the same features? The answer matters. I think the honest

answer is that the LLM can process the full player roster as a contextual narrative rather than requiring manual feature engineering — it implicitly captures interactions between players, tactical matchups, and form trajectories that would be hard to encode as tabular features. But you should also acknowledge that a simpler model might perform just as well or better, and that comparing LLM vs. traditional ML baselines would be a natural extension.

What's your take on these three points? And is there anything about the logic chain that doesn't feel right to you?

1. we are not just evaluating the score and match event, the llm also got better fine-tuned on reasoning behind each prediction. And to beat a real bookmaker model is not necessary for this project since the performance is not part of the grading schema.
2. I have all the leagues and world cup data, but we are targeting at world cup first. Maybe we can back test other 2022 leagues game later that's definitely doable.
3. Yes. and I think also most importantly LLM can have transparent reasoning behind each of its prediction, instead of other ML models which is simply a black box and there are no ways for users to justify the prediction.

All three points are sharp. Let me refine the logic based on your corrections.

On point 1 — reasoning quality as a first-class output. This is actually the strongest differentiator of the whole strategy and I undersold it. The model doesn't just output "Argentina 72%, Saudi Arabia 18%, Draw 10%." It outputs *why* — "Argentina's midfield dominance is driven by Player X's 89% pass completion rate in La Liga combined with Player Y's 2.3 key passes per 90, but Saudi Arabia's compact defensive structure in AFC qualifiers (0.8 goals conceded per match) could neutralize this." That reasoning chain is itself valuable because it makes the prediction auditable. A quant trader can read the reasoning and decide whether the logic is sound before placing the bet. This maps directly to Professor Yang's core principle of **measurable outcomes** — you're not just measuring whether the prediction was right, you're measuring whether the *reasoning* was sound.

On point 2 — backtest scope. Good. World Cup 2022 as the primary backtest (64 matches, clean ground truth), with league 2022 data as an expansion for robustness testing. That gives you a two-tier validation story: narrow but clean (WC) and broad but noisier (leagues).

On point 3 — interpretability as alpha. This is the killer argument and you should lead with it in the essay. Traditional ML models for sports prediction (Elo, XGBoost, neural nets) are black boxes — they give you a number but no justification. Bookmaker odds are even worse: you don't know *why* they priced a match at those odds. Your LLM provides transparent, natural-language reasoning grounded in specific player statistics. This means three things for the strategy:

First, the trader can **audit** the signal before acting — if the model's reasoning cites a player who's actually injured, you know to override. Second, you can **debug** bad predictions post-hoc by

examining the reasoning chain, which is impossible with XGBoost. Third, the reasoning itself becomes a **feedback signal** for model improvement — you can identify systematic reasoning failures (like consistently overweighting offensive stats) and address them in the next fine-tuning iteration.

So the refined strategy logic is:

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Data (player stats across leagues/WC)
  → Fine-tuned LLM (probability + reasoning)
    → Edge detection (model prob vs. bookmaker implied prob)
      → Reasoning audit (human reviews LLM justification)
        → Position sizing (fractional Kelly)
          → Execution (place bet)
            → Post-match review (outcome + reasoning quality)
```

The reasoning audit step is what makes this a **human-in-the-loop quant strategy** rather than a fully automated black box. That's a legitimate design choice you can defend — it mirrors how many real quant firms operate, where systematic signals are generated algorithmically but portfolio managers retain discretion to override.

One more thing I want to nail down before the essay. The assignment asks you to illustrate your **pattern** and **model**. In the language of slide 18:

- Your **pattern** is Statistical (ML-based, testable)
- But there's also a **Behavioral** element — you're exploiting the fact that betting markets for World Cup matches are influenced by narrative bias, national sentiment, and recency effects. Public bettors overweight famous team names and recent headlines. Your model, trained on granular player data, is less susceptible to these biases.

Do you want to frame it as purely statistical, or lean into the behavioral mispricing angle too? I think the dual framing (statistical model exploiting behavioral mispricing) is stronger and more aligned with what the lecture teaches, but it's your call.

Also — do you have access to historical bookmaker odds for World Cup 2022? That would make the backtest much more concrete. If not, we can still frame the strategy conceptually and note that historical odds data would be needed for a full backtest.

I think the dual framing is good. And I don't have historical bookmaker odds. we just need to compare with ourselves meaning cal the performance or accuracy. we are letting the model predict the score and match event, with reasoning for justification, and compare with the 2022 world cup real match results as the ground truth. why do we need histrocial bookmaker odds?

You're right — I was overcomplicating it. Let me recalibrate.

The bookmaker odds angle only matters if you're framing this as "beat the market." But that's not what you're doing. You're framing this as: **I built a fine-tuned LLM that predicts World Cup match outcomes with transparent reasoning, and I evaluate it against actual results.** The "trading strategy" framing comes from the structure and discipline around the prediction pipeline, not from literally arbitraging a bookmaker.

So the backtest is straightforward: your model predicts match outcomes (score, win/draw/loss, match events) with reasoning → you compare against the actual 2022 World Cup results → you measure accuracy, calibration, and reasoning quality. That's the eval.

The bookmaker comparison would be a nice-to-have extension — "our model's predicted probabilities were better calibrated than bookmaker implied probabilities" — but it's not necessary for the assignment, and you shouldn't introduce a dependency you don't have data for.

Revised strategy evaluation framework (no bookmaker data needed):

The metrics you'd report are prediction accuracy (did the model get win/draw/loss right), score prediction accuracy (how close was the predicted scoreline), calibration (when the model says 70% win probability, does that team win roughly 70% of the time — though small sample caveat applies), reasoning quality (does the justification cite relevant player stats and make logically sound arguments, even when the prediction is wrong), and upset detection (did the model correctly identify any of the famous 2022 upsets like Saudi Arabia beating Argentina or Japan beating Germany, and if so, what reasoning did it give).

That last one is actually the most interesting for the essay. The 2022 World Cup was famous for upsets. If your model's reasoning can explain *why* an upset was plausible — even if the model still predicted the favorite to win — that demonstrates the value of interpretable predictions over black-box confidence scores.

Now I think we have the full picture. The strategy logic is clean:

The **market** is World Cup match outcomes. The **signal** is a fine-tuned LLaMA model that ingests player-level statistics from club leagues and international tournaments and outputs match predictions with natural-language reasoning. The **pattern** is statistical (ML-based) exploiting behavioral tendencies (narrative bias in how people evaluate national teams). The **core innovation** is that the LLM provides transparent, auditable reasoning chains — not just probabilities — enabling human-in-the-loop validation before and diagnostic analysis after each prediction. The **evaluation** is against ground truth match results from 2022 World Cup, measuring prediction accuracy, score proximity, and reasoning quality.

Ready for me to draft the essay now? Or is there anything else about the logic you want to tighten?

Good. now let's finish the table 1, following the requirement "Reorganize your trading idea with Table 1 (Core Logic Design of Trading Strategies)". see the screenshot for the table. I think I will simply use docx for the table.

Let me look at the table structure carefully. It has checkboxes for the middle rows, meaning you pick one from each pair (left or right column), and then fill in the description in the third column.

Let me think through each row for your strategy:

Targeting Market: FIFA World Cup match outcome betting markets (specifically, pre-match win/draw/loss wagers)

Row 2 - Phenomena Description vs Theoretical Framework: I'd check **Phenomena Description** since your starting point is an observed phenomenon: betting markets for World Cup matches are disproportionately influenced by team reputation, media narratives, and casual bettor sentiment, leading to systematic mispricings especially around upsets.

Row 3 - Analytical Framework vs Theoretical Hypothesis: I'd check **Analytical Framework** since you built a concrete analytical tool (fine-tuned LLM) rather than starting from a theoretical hypothesis. The framework is: ingest player-level stats across leagues/tournaments → generate match predictions with reasoning → compare against market pricing.

Row 4 - Phenomena Interpretation vs Theoretical Reasoning: I'd check **Phenomena Interpretation** since your interpretation of *why* the phenomenon exists is behavioral: World Cup matches attract high volumes of sentiment-driven bettors, the matches happen infrequently so historical data is thin, and national pride distorts risk assessment.

Trading Opportunity: When the model's predicted probability for an outcome diverges from the market's implied probability by more than a confidence threshold, that gap represents exploitable alpha.

Trading Operations: Pre-match only. For each World Cup match, generate prediction with reasoning chain → human reviews reasoning for factual errors (injuries, suspensions) → if edge exceeds threshold, size position via fractional Kelly criterion → place bet before kickoff. Hard cap per match, drawdown circuit breaker on cumulative losses.

Financial Instrument(s): Fixed-odds sports betting contracts (win/draw/loss) on regulated sportsbooks. Could extend to Asian handicap or over/under markets.

Does this mapping feel right to you? Specifically, do you agree with checking the left column for all three middle rows (Phenomena Description, Analytical Framework, Phenomena Interpretation)? Or do you think any of those should be the right column instead? Once you confirm, I'll generate the docx.